

ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY & SCIENCES

Autonomous Institution Affiliated to Andhra University | Approved by AICTE

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Course: Full Stack Web Development Lab

Observation Task: #02

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1. TITLE OF THE EXPERIMENT

Design of a Styled Webpage with CSS3 Flexbox Layout and HTML5 Canvas Graphics

2. AIM

To design a modern styled webpage applying external CSS3 gradients, navigation bar hover effects, flexbox cards, and dynamically drawing 2D shapes (rectangles, arcs, lines, and text) on an HTML5 canvas using JavaScript.

3. PROCEDURE / ALGORITHM

Step 1: Create an HTML5 page linked to an external stylesheet (style.css).

Step 2: Design a gradient header and centered navigation bar with interactive yellow hover effects.

Step 3: Implement a responsive card container utilizing CSS3 Flexbox (display: flex, gap: 20px).

Step 4: Define an HTML5 <canvas id='myCanvas'> element with width and height attributes.

Step 5: Write JavaScript code to acquire the 2D rendering context (canvas.getContext('2d')) and draw shapes.

Step 6: Render a filled rectangle (fillRect), stroked circle (arc), line (moveTo/lineTo), and canvas text (fillText).

4. TEST CASES / VERIFICATION

Test Case	Description / Action	Expected Output	Status
TC-01	CSS3 Header & Nav Styling	Gradient background and navigation link hover effects	Passed
TC-02	Flexbox Card Container	Cards aligned horizontally with consistent gap and box shadow	Passed
TC-03	Canvas 2D Context	Obtain 2D context from DOM element without errors	Passed
TC-04	Canvas Shape Rendering	Rectangle, circle, line, and text 'sai' rendered accurately	Passed

5. LEARNING OUTCOME / RESULT

The styled webpage with external CSS3 and HTML5 Canvas was successfully implemented and verified, displaying the styled flexbox cards and dynamic 2D canvas graphics.

6. SOURCE CODE

File: index.html

```
<!DOCTYPE html>
<html>
<head>
<title>My Styled Webpage</title>
<link rel="stylesheet" href="style.css">
</head>

<body>

<header>
<h1>My Styled Webpage</h1>
</header>

<nav>
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Contact</a>
</nav>

<canvas id="myCanvas" width="500" height="220"></canvas>

<section class="container">

<div class="card">
<h2>HTML5</h2>
<p>HTML is used to create web pages.</p>
<button>Read More</button>
</div>

<div class="card">
<h2>CSS3</h2>
<p>CSS is used to style web pages.</p>
<button>Read More</button>
</div>

<div class="card">
<h2>JavaScript</h2>
<p>JavaScript adds interactivity.</p>
<button>Read More</button>
</div>

</section>

<footer>
<p>© 2026 ANITS</p>
</footer>

<script>

let canvas=document.getElementById("myCanvas");
let ctx=canvas.getContext("2d");

ctx.fillRect(20,20,120,70);

ctx.beginPath();
ctx.arc(220,55,35,0,2*Math.PI);
ctx.stroke();

ctx.beginPath();
ctx.moveTo(20,130);
ctx.lineTo(300,130);
ctx.stroke();

ctx.font="28px Arial";
ctx.fillText("sai",170,190);

</script>

</body>
```

</html>

File: style.css

```
body{
  margin:0;
  font-family:Arial,sans-serif;
  background:#f2f2f2;
}

header{
  background:linear-gradient(to right,#0066cc,#8000ff);
  color:white;
  text-align:center;
  padding:20px;
  text-shadow:2px 2px 5px gray;
}

nav{
  background:#333;
  text-align:center;
  padding:15px;
}

nav a{
  color:white;
  text-decoration:none;
  margin:20px;
  font-size:18px;
}

nav a:hover{
  color:yellow;
}

.container{
  display:flex;
  justify-content:center;
  gap:20px;
  margin:30px;
}

.card{
  background:white;
  width:250px;
  padding:20px;
  border:2px solid gray;
  border-radius:15px;
  text-align:center;
  box-shadow:5px 5px 10px gray;
}

button{
  background:green;
  color:white;
  border:none;
  padding:10px 15px;
  border-radius:5px;
  cursor:pointer;
}

button:hover{
  background:red;
}

footer{
  background:black;
  color:white;
  text-align:center;
  padding:15px;
}
```

```
}    margin-top:20px;
```

7. OUTPUT SCREENSHOTS

Output Screenshot: Screenshot 2026-08-22 194021.png

